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Build a Carbon Credit Management System

Note: This task is designed to simulate real-world development where you'll encounter new technologies and need to research solutions independently. It's an opportunity to demonstrate not just your coding ability, but also your learning process and professional development practices

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Important Note

This project provides comprehensive documentation and specifications in a completely real-world style because carbon offsetting is an actual global business challenge. We've included:

- **Complete API specifications** with Swagger/OpenAPI documentation
- **Sample UI mockups** showing the entire user journey
- **Database schemas** with both SQL and NoSQL options
- **Business context** explaining how carbon credits work in practice
- **Sample data structures** for all operations
  - Everything is designed to give you the full context a developer would receive in a real job. Feel free to:
  - Fine-tune the specifications based on your understanding
  - Add improvements or optimizations
  - Make architectural decisions that make sense
  - Implement additional features you think would add value

The goal is to see how you approach a real-world problem with real business requirements, not just a coding exercise. Treat this as if you're building an MVP for an actual startup in the sustainability space.

Remember: Good developers don't just code - they understand the business problem and create solutions that work for real users.

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What is Carbon Offsetting?

- **Definition**: Compensating for CO_2 emissions by funding projects that remove or reduce greenhouse gases from the atmosphere
- **How it works**: Calculate emissions → Purchase credits → Fund green projects (1 credit = 1 ton CO_2 removed)
- **Common offset projects**: Reforestation, renewable energy, carbon capture, ocean cleanup, methane reduction

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Carbon Offsetting in Business

- **Corporate responsibility**: Companies achieve carbon neutrality by offsetting unavoidable emissions
- **Compliance & regulation**: Meeting government mandates and industry standards for emissions
- **Brand value**: Demonstrating environmental commitment to customers and stakeholders
- **Supply chain requirements**: Many large corporations require suppliers to track and offset emissions
- **Carbon markets**: Trading credits creates financial incentives for emission reduction

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What This Project Does

- **Tracks carbon footprint**: Measures emissions from business activities (transport, energy, manufacturing)
- **Rewards green actions**: Converts reduction activities into tradeable carbon credits
- **Marketplace for offsets**: Connects businesses with verified carbon reduction projects
- **Transparent ledger**: Blockchain-inspired system ensures credits aren't double-counted
- **Simplifies compliance**: Automated reporting and credit management for businesses
- **Encourages participation**: Gamifies sustainability with credit sharing and rewards

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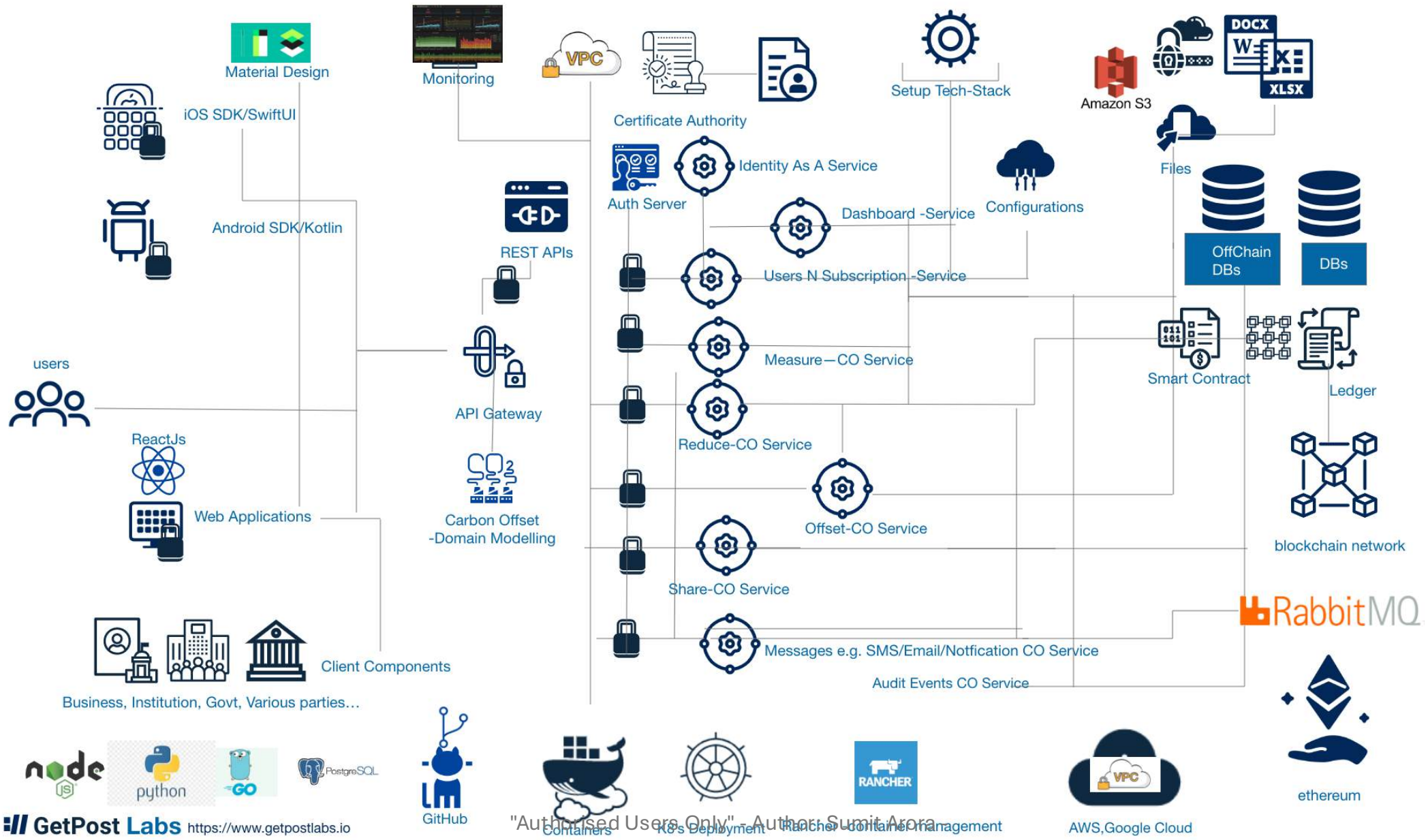
Architecture

Note: Focus on the core business logic, not blockchain complexity. Use a regular database to simulate blockchain storage.

Architecture

A FLAT VIEW OF BLOCKCHAIN-ENABLED CARBON OFFSETTING SYSTEM



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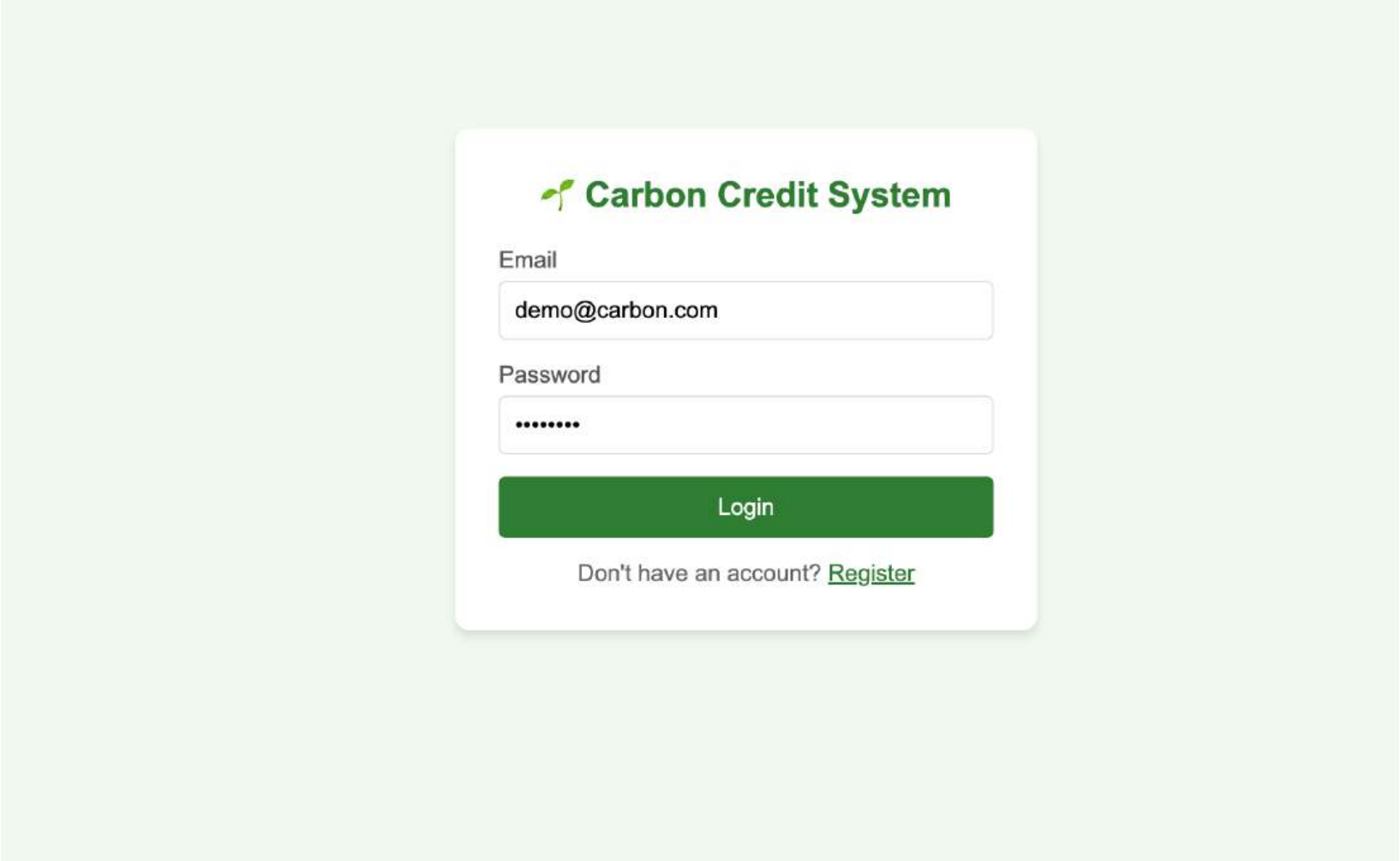
UI – Refer - carbon-credit-ui.html


Carbon Credit System
Email
demo@carbon.com
Password
••••••••
Login
Don't have an account? Register

Create Account

Full Name

John Doe

Email

user@example.com

Password

••••••••

Confirm Password

••••••••

Register

Already have an account? Login

🌱 Carbon Credit Dashboard

Welcome, John Doe Logout

Your Carbon Credit Balance

2,450

Credits Available

Total Emissions	Total Reductions	Credits Purchased	Credits Shared
8,320 kg	5,870 kg	1,500	250

Quick Actions

📊 **Measure**
Track your carbon emissions

🌿 **Reduce**
Log reduction activities

💚 **Offset**
Purchase carbon credits

🤝 **Share**
Transfer credits to others

Recent Transactions

Date	Type	Description	Amount	Balance
2025-08-03	Reduce	Solar panel installation	+500 credits	2,450
2025-08-02	Measure	Car travel - 150km	-35 kg CO_2	1,950
2025-08-01	Offset	Amazon Rainforest Project	+1,000 credits	1,985
2025-07-30	Share	Transfer to jane@example.com	-100 credits	985

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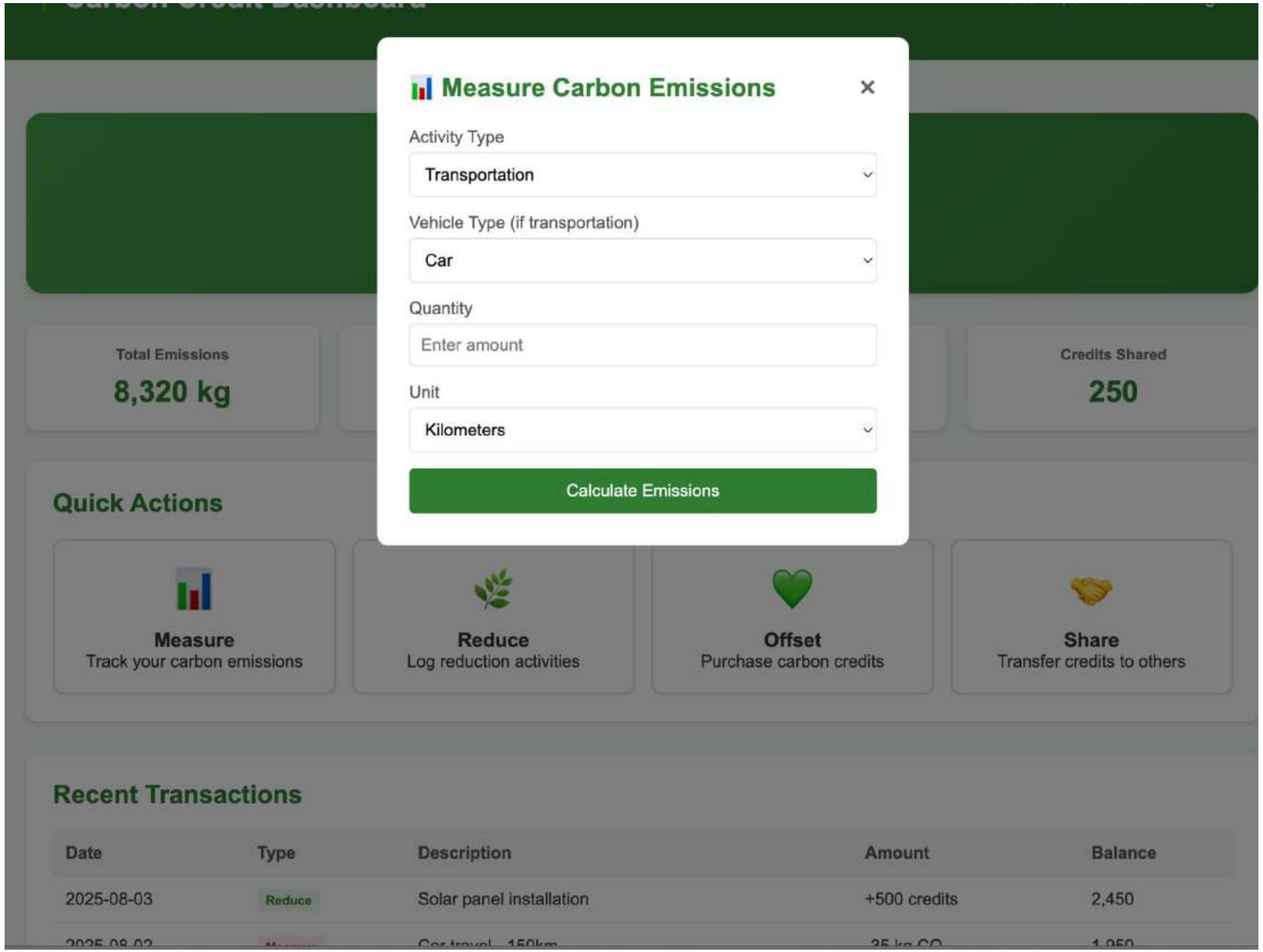

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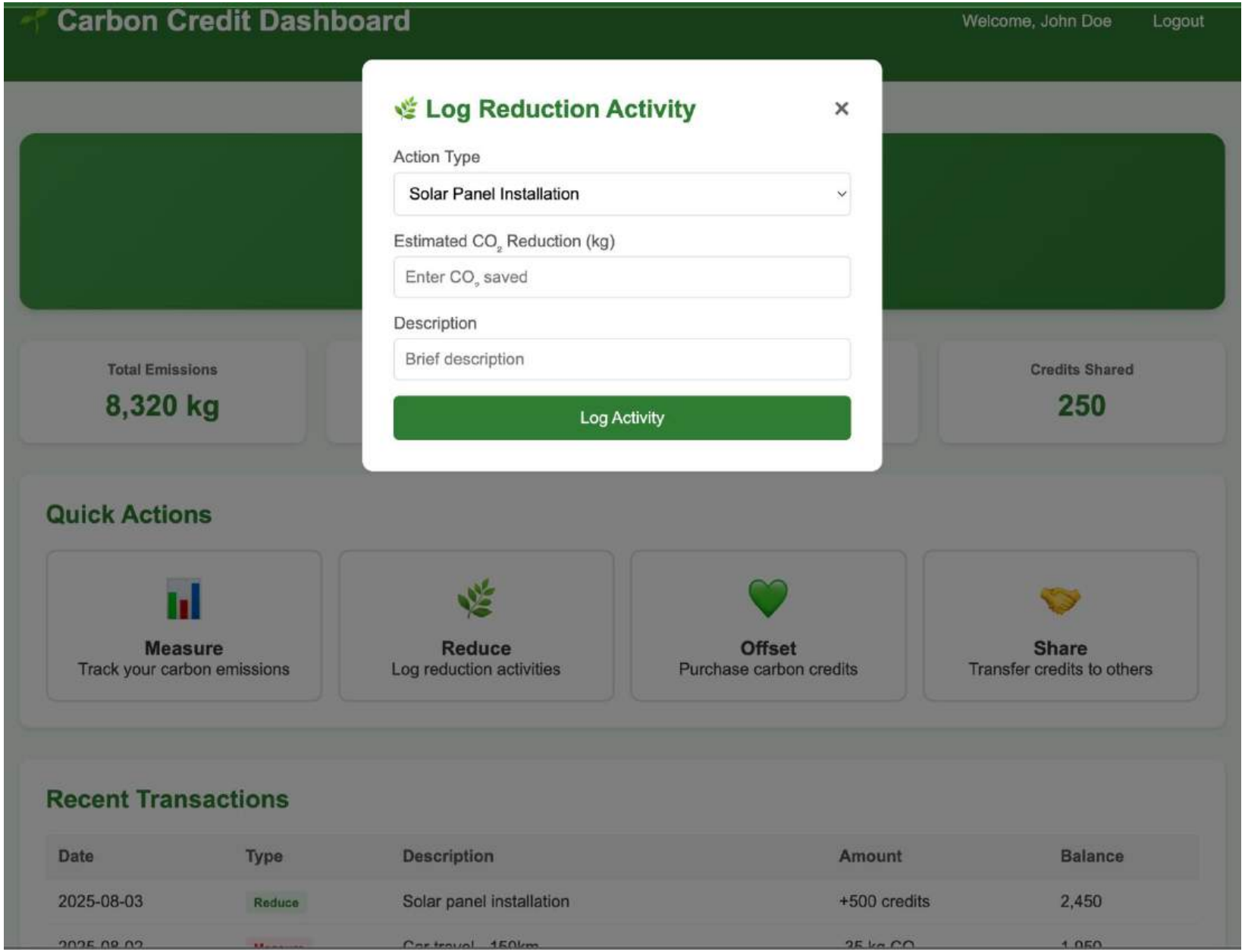

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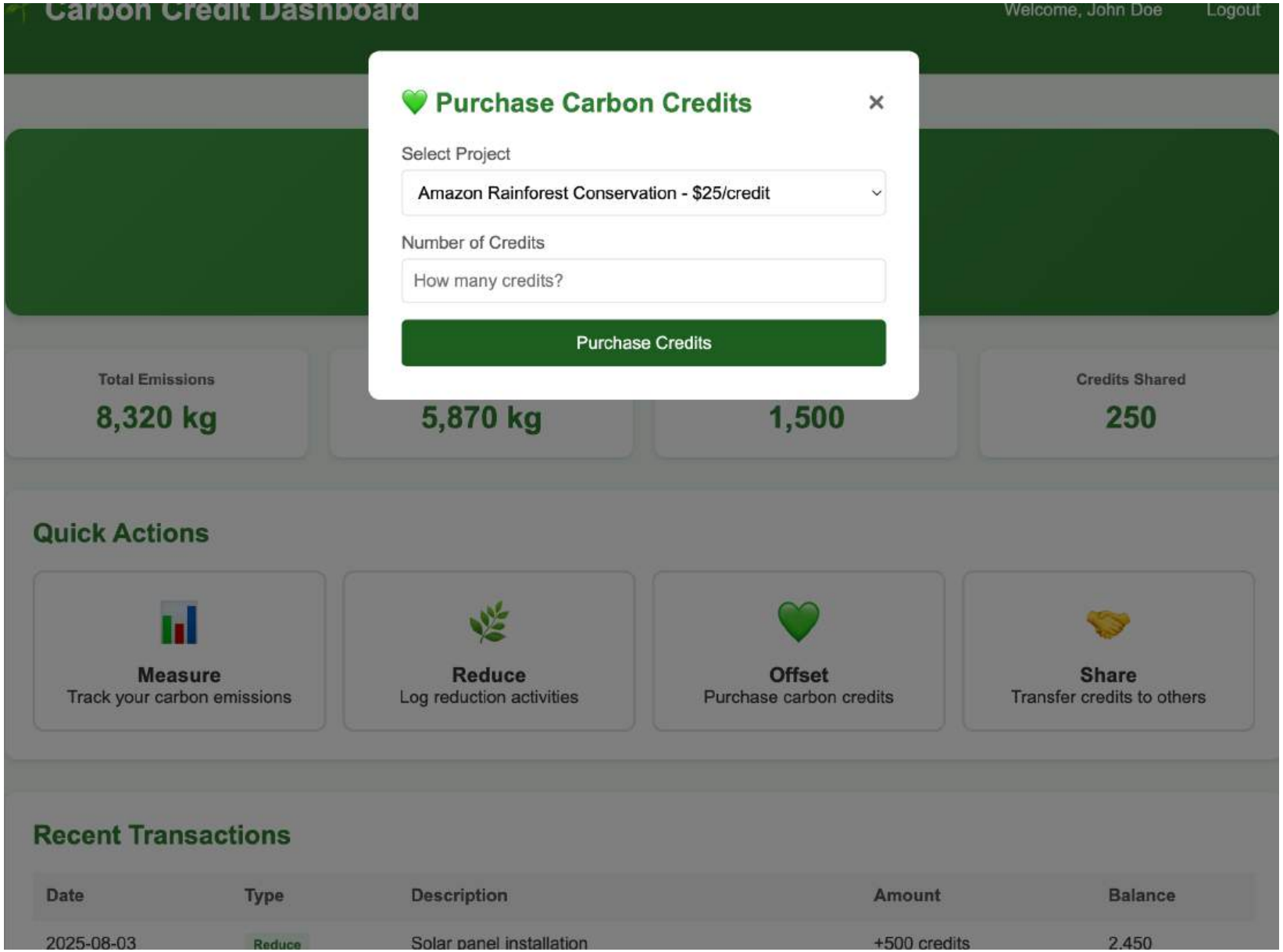

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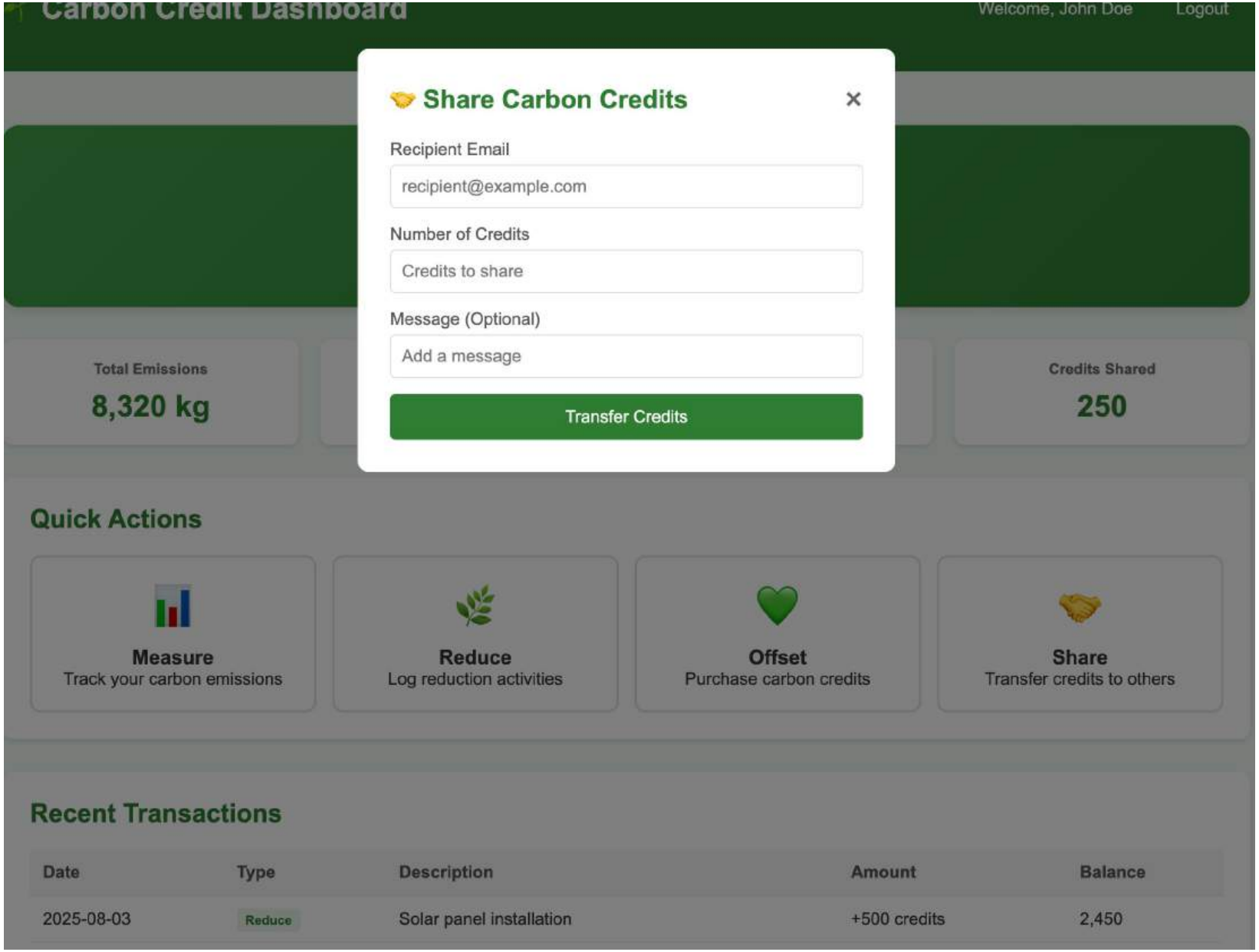
Carbon Credit Dashboard
Welcome, John Doe
Logout
Share Carbon Credits
Recipient Email
recipient@example.com
Number of Credits
Credits to share
Message (Optional)
Add a message
Transfer Credits
Total Emissions
8,320 kg
Credits Shared
250
Quick Actions
Measure
Track your carbon emissions
Reduce
Log reduction activities
Offset
Purchase carbon credits
Share
Transfer credits to others
Recent Transactions
Date
Type
Description
Amount
Balance
2025-08-03
Reduce
Solar panel installation
+500 credits
2,450

API – Refer - carbon-api-swagger.yaml

Carbon Credit Management API 1.0.0 OAS 3.0

API for managing carbon credits - measure, reduce, offset, and share

Servers

http://localhost:8080/api - Development server

Authorize

Authentication

POST	/auth/register	Register new user
POST	/auth/login	Login user

Carbon Operations

POST	/carbon/measure	Record carbon emission activity
GET	/carbon/measure	Get user's carbon emission history
POST	/carbon/reduce	Record carbon reduction action
POST	/carbon/offset	Purchase carbon offset credits
GET	/carbon/offset/projects	Get available offset projects
POST	/carbon/share	Share carbon credits with another user

Dashboard

GET	/dashboard/summary	Get user dashboard summary

DB - carbon-database-schema.sql

APPENDIX

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References

- https://offsetguide.org/
- https://interactive.carbonbrief.org/carbon-offsets-2023/index.html